PRODUCT USER MANUAL

For Multi-Year WAVE Products

WAVE_GLO_PHY_SWH_L3_MY_014_005

WAVE_GLO_PHY_SPC_L3_MY_014_006

WAVE_GLO_PHY_SWH_L4_MY_014_007

Issue: 3.0

Contributors : Romain Husson (CLS), Elodie Charles (CLS), Cécile Béchonnet (CLS)

Approval Date : NOVEMBER 2024

MERCATOR OCEAN
INTERNATIONAL
2avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2021/06/11	All	First version of document at CMEMS for the WAVE_GLO_PHY_SPC_L3_MY_014_006	R. Husson, E. Charles	E. Charles
2.0	2021/09/10	All	Integration of new multi-year products: WAVE_GLO_PHY_SWH_L3_MY_014_005, WAVE_GLO_PHY_SWH_L4_MY_014_007	R. Husson, E. Charles	E. Charles
3.0	2024/04/22	All	New mapping method instantaneous SWH field in L4 SWH product	C.Béchonnet	P.Zunino

Table of Contents

GLOSSARY AND ABBREVIATIONS *4*

DATA ACCESS **6**

I INTRODUCTION *7*

I.1 Summary **7**

I.2 History of changes **7**

II DESCRIPTION OF THE PRODUCT SPECIFICATION *8*

II.1 General Information **8**

II.2 Details of datasets **11**

II.3 Production System Description **13**

II.3.1 WAVE-TAC SWH system 13

II.3.2 WAVE-TAC SPC system 15

II.4 Processing information **16**

II.4.1 Update Time 16

II.4.2 Temporal extent of data stored on delivery mechanism 16

II.4.3 Time averaging 16

III FILE FORMAT *17*

IV FILES NOMENCLATURE *18*

IV.1 Nomenclature of files when downloaded through the CMEMS Web Portal Subset Service **18**

IV.2 Nomenclature of original files **18**

IV.3 Other information: mean centre of Products, land mask value, missing value **20**

IV.4 File size **20**

IV.5 Structure of files **21**

V REFERENCES *22*

VI ANNEX *23*

VI.1 WAVE_GLO_PHY_SWH_L3_MY_014_005 files **23**

VI.1.1 WAVE_GLO_PHY_SPC_L3_MY_014_006 files 24

VI.1.2 WAVE_GLO_PHY_SWH_L4_MY_014_007 files 33

GLOSSARY AND ABBREVIATIONS

ADD	Architecture and Design Document
AL	Saral/AltiKa
C2	Cryosat-2
CalVal	Calibration/Validation
CF	Climate Forecast (convention for NetCDF)
CFO	CFOSAT
CLS	Collecte Localisation Satellite
CMEMS	Copernicus Marine Environment Monitoring Service
DGF	Direct Get File (FTP-like CMEMS service tool to download a NetCDF file)
DT	Delayed Time
DU	Dissemination Unit
GDR	Geophysical Data Record
GOP	Geophysical Ocean Product
HY-2B	Hai Yang 2B
IGDR	Intermediate Geophysical Data Record
J3	Jason-3
KPI	Key Production Indicator
L2/L2P/L3/L4	Level 2 / Level 2 Plus / Level 3 / Level 4
LRM	Low Resolution Mode
MY	Multi-Year
NetCDF	Network Common Data Form
NOP	NRT Ocean Product
NRT	Near Real Time
OE	Orbit Error

OGDR	Operational Geophysical data record
OLA	Operational Level Agreement
PIT	Product Information Table
PLRM	Pseudo LRM
PQ	Product Quality
PU	Production Unit
PUM	Product User Manual
QUID	QUality Information Document
RMS	Root Mean Square
RPC	Remote Protocol Client
RT	Real Time
S1A	Sentinel-1A
S1B	Sentinel-1B
S3A	Sentinel-3A
S3B	Sentinel-3B
SAR	Synthetic Aperture Radar
SARIn	SAR Interferometric
SPC	SPeCtral
SRD	System Requirement Document
SWH	Significant Wave Height
TAC	Thematic Assembly Center
TDS	Thredds Data Server
VAVH	Visual AVerage Height
WMS	Web Map Service

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

This document is the user manual for the CMEMS (Copernicus Marine Environment Monitoring Service) Multi-Year satellite wave products produced by the WAVE-TAC (Thematic Assembly Centre).

I.1 Summary

The following multi-year along-track and gridded products are delivered by the WAVE-TAC and are detailed in this user manual:

WAVE_GLO_PHY_SWH_L3_MY_014_005: Multi-year altimetry along-track significant wave height level-3 product based on Jason-1, Jason-2, Jason-3, Envisat, Saral/AltiKa, Cryosat-2 and CFOSAT nadir measurements. This product is based on the output of the ESA Climate Change Initiative Sea State project and on the reprocessing of CFOSAT Level-2P products (CNES/CLS).

WAVE_GLO_PHY_SPC_L3_MY_014_006: Multi-year SAR spectral integral parameter level-3 product based on Sentinel-1A and Sentinel-1B measurements.

WAVE_GLO_PHY_SWH_L4_MY_014_007: Multi-year gridded significant wave height level-4 product based on all available significant wave height multi-year level-3 products.

For each along-track product, one dataset is available for each satellite mission.

I.2 History of changes

Date	Description of changes and impacted product
JUN-2021	Launch of a new multi-year wave product: WAVE_GLO_PHY_SPC_L3_MY_014_006
DEC-2021	Launch of three new multi-year wave products: WAVE_GLO_PHY_SWH_L3_MY_014_005, WAVE_GLO_PHY_SWH_L4_MY_014_007
APR-2024	Update product WAVE_GLO_PHY_SWH_L4_NRT_014_003: new mapping method for the instantaneous swh field.

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

Product Lines	**WAVE_GLO_PHY_SWH_L3_MY_014_005**
Geographical coverage	Global
Variables	**VAVH**: sea surface significant wave height, i.e. the mean wave height of the highest third of the waves ($H_{1/3}$) **VAVH_UNFILTERED**: sea surface significant wave height, i.e. the mean wave height of the highest third of the waves ($H_{1/3}$), before applying the along-track denoising
Product Type	N/A
Available time series	See QUID
Temporal resolution	N/A
Target delivery time	Instantaneous (one file contains 3 hours of measurements)
Delivery mechanism	**Marine Data Store**
Horizontal resolution	7km (along-track)
Number of vertical levels	Surface only
Format	NetCDF-4

Table 1: WAVE-TAC WAVE_GLO_PHY_SWH_L3_MY_014_005 product

Product name	**WAVE_GLO_PHY_SPC_L3_MY_014_006**
Geographic coverage	Global (except European coasts)
Variables	The following integral parameters are used to describe observed swell fields along propagation time, from the storm source to the land:

	VAVH: sea surface significant wave height, i.e. the mean wave height of the highest third of the waves ($H_{1/3}$) for the considered wave partition **VTPK**: sea surface wave period at partition variance spectral density maximum, i.e. wave period at partition spectral peak / peak period (Tp) **VPED**: sea surface wave from direction at partition spectral peak, i.e. wave principal direction at partition spectral peak (Dp) Where invalid, these variables are replaced by their fill values. Additional information at observation level are also available in a separate group untitled "**obs_params**". Additional information for more expert users are also available in a separate group untitled "**exp_params**". **Additional format:** This dataset is available in two different formats: - Files gathering only waves measurements belonging to the same swell field, covering vaious time periods of 10 to 15 days, - Files gathering only waves measurements belonging to the all swell fields, but covering covering a 3 hour time period, Those two file formats contain the same information but arranged in two different ways in order to best satisfy all user needs.		
Update frequency	6 months		
Available time series	See QUID		
Target delivery time			
Temporal resolution	3-hourly: propagated swell observations are provided with a 3-hour time step from one storm source until dissipation offshore or at the coast		
Delivery mechanisms	**Marine Data Store**		
Horizontal resolution	Discrete		
Number of vertical levels	Surface only		
Format	NetCDF-4		

Table 2: WAVE-TAC WAVE_GLO_PHY_SPC_L3_MY_014_006 product

Product name	**WAVE_GLO_PHY_SWH_L4_MY_014_007**		
Geographic coverage	Global		
Variables	**VAVH_DAILY_MEAN**: daily mean significant wave height **VAVH_DAILY_STD**: daily standard deviation of significant wave height **VAVH_DAILY_MAX**: daily maximum significant wave height **VAVH_DAILY_NBR**: daily number of significant wave height measurements **VAVH_INST**: instantaneous significant wave height estimated from all available satellite measurements, accounting for their spatial and temporal proximity**.** **VAVH_INST_FLAG**: flag associated with a post-processing modification of the data (threshold set to 0 m). Where the value is invalid, the flag is set to 1, otherwise to 0.		
Update frequency	N/A		
Available time series	See QUID		
Target delivery time	N/A		
Temporal resolution	Daily		
Delivery mechanisms	**Marine Data Store**		
Horizontal resolution	2°x 2° for DAILY_MEAN and 0.5°x0.5° for INST		
Number of vertical levels	Surface only		

Format	NetCDF-4

Table 3: WAVE-TAC WAVE_GLO_PHY_SWH_L4_MY_014_007 product

Detailed information on the systems and products are on CMEMS web site: marine.copernicus.eu

II.2 Details of datasets

WAVE_GLO_PHY_SWH_L3_MY_014_005
Datasets (one per mission): cci_obs-wave_glo_phy-swh_my_al-l3_PT1S cci_obs-wave_glo_phy-swh_my_c2-l3_PT1S cci_obs-wave_glo_phy-swh_my_en-l3_PT1S cci_obs-wave_glo_phy-swh_my_j1-l3_PT1S cci_obs-wave_glo_phy-swh_my_j2-l3_PT1S cci_obs-wave_glo_phy-swh_my_j3-l3_PT1S cmems_obs-wave_glo_phy-swh_my_cfo-l3_PT1S **Variables name in the NetCDF file and Unit: Long_name & Standard_name**
VAVH [m] Significant wave height (H1/3) on main altimeter frequency band sea_surface_wave_significant_height
VAVH_UNFILTERED [m] Significant wave height (H1/3) on main altimeter frequency band sea_surface_wave_significant_height

Table 3: list of the datasets and variable names and unit for the WAVE_GLO_PHY_SWH_L3_MY_014_005 product

WAVE_GLO_PHY_SPC_L3_MY_014_006
Datasets: Dataset-wav-sar-l3-spc-rep-global-* **Variables name in the NetCDF file and Unit: Long_name & Standard_name**
VAVH [m] sea_surface_swell_wave_part_significant_height And associated flags obs_params/VAVH_flag Flag for the sea_surface_swell_wave_part_significant_height
VTPK [s] Sea_surface_swell_wave_part_peak_period And associated flags obs_param/VTPK_flag

Flag for the sea_surface_swell_wave_part_peak_period
VPED [deg] sea_surface_swell_wave_part_princ_from_direction obs_params/VPED_flag Flag for the sea_surface_swell_wave_part_princ_from_direction

Table 4: list of the datasets and variable names and unit for the WAVE_GLO_PHY_SPC_L3_MY_014_006 product

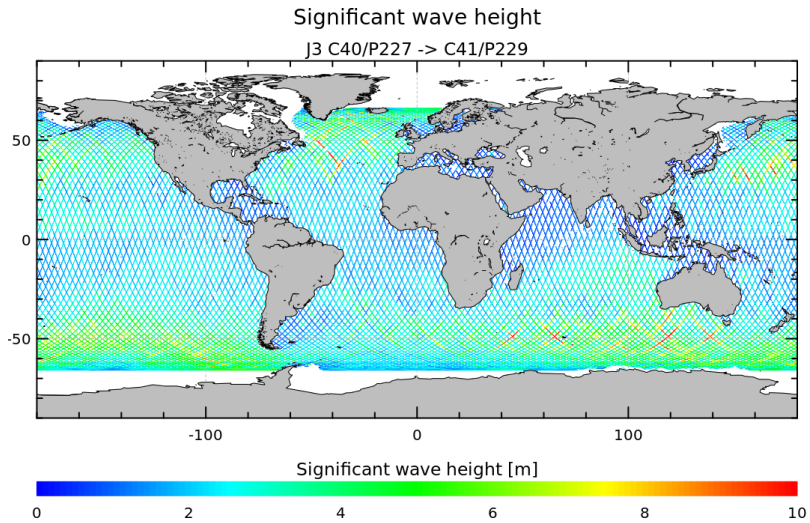
WAVE_GLO_PHY_SWH_L4_MY_014_007
Datasets (one for daily mean variables (P1D-m), other for instantaneous variables (P1D-i): **cmems_obs-wave_phy-swh_my_multi_l4_2deg_P1D-m** **cmems_obs-wave_phy-swh_my_multi_l4_0.5deg_P1D-i** **Variables name in the NetCDF file and Unit: Long_name & Standard_name**
VAVH_DAILY_MEAN [m] sea_surface_wave_significant_height_daily_mean
VAVH_DAILY_STD [m] sea_surface_significant_height_daily_standard_deviation
VAVH_DAILY_MAX [m] sea_surface_significant_height_daily_maximum
VAVH_DAILY_NBR [-] sea_surface_significant_height_daily_number_of_observations
VAVH_INST [m] sea_surface_significant_height
VAVH_INST_FLAG [-] Flag associated with the modification of VAVH_INST variable sea_surface_wave_significant_height_flag

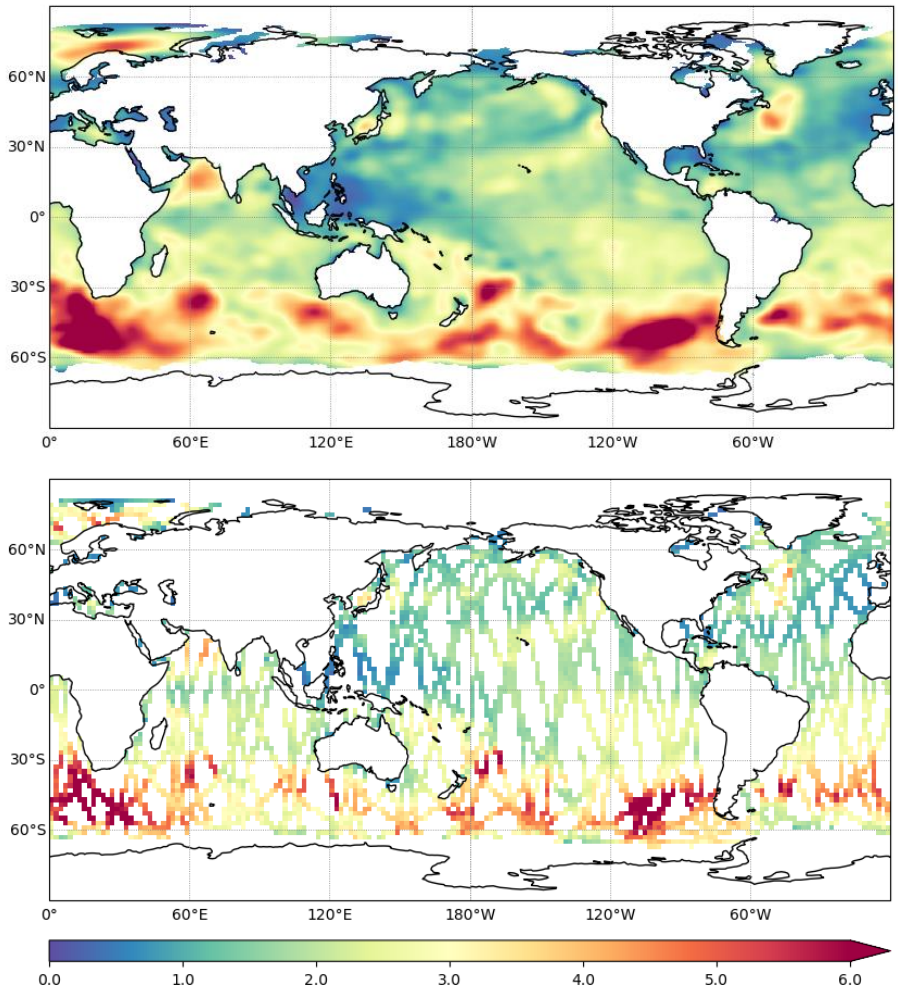
Table 5: list of the datasets and variable names and unit for the WAVE_GLO_PHY_SWH_L4_MY_014_007 product

Note for users: The VAVH variable stands for Visual AVerage Heigh of the highest 1/3 wave (H1/3). Significant wave height is a statistic computed from wave measurements and corresponds to the average height of the highest one third of the waves, where the height is defined as the vertical distance from a wave trough to the following wave crest. The first 'V' in VAVH comes from the in-situ community.

II.3 Production System Description

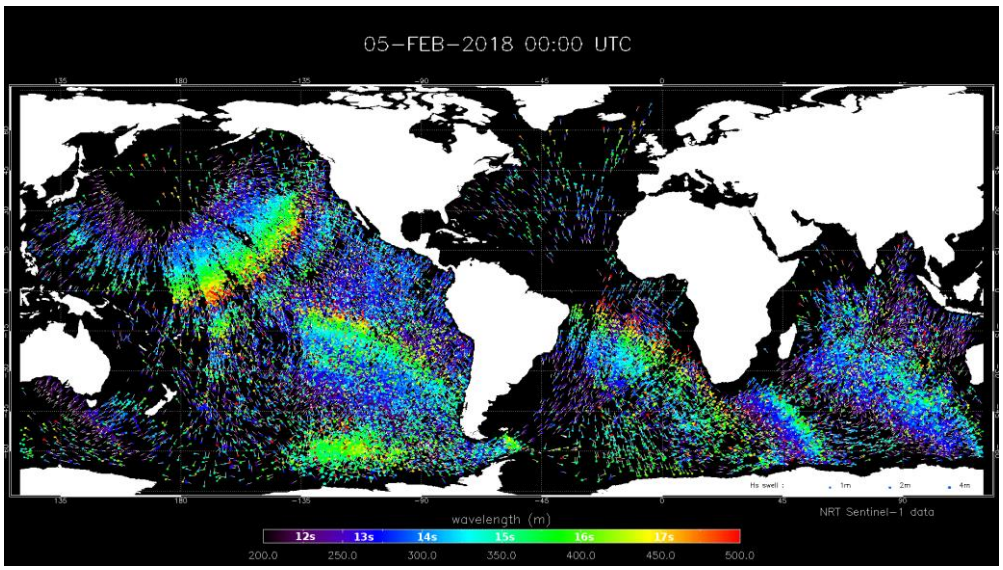
II.3.1 WAVE-TAC SWH system

WAVE-TAC SWH SYSTEM	
Processed products	WAVE_GLO_PHY_SWH_L3_MY_014_005
Domain Resolution and grid Geographic coverage	The altimeter measurements are delivered over global Ocean. The measurements are made along-track at the surface with a 7-km resolution. For Jason-1, 2 and 3, the coverage is maximum 66°S/66°N in latitude; for SARAL/AltiKa, 81.5°S/81.5°N; for Cryosat-2, 88°S/88°N; and for CFOSAT, 83°S/83°N. See QUID for details.  Figure 1*: Example of wave measurements from along-track Jason-3 altimeter. One point every 7km along the satellite passes. The coverage is 10 days [pass 227 cycle 40 to pass 229 cycle 41 (one cycle is 10 days)] for Jason-3.*
Processed products	WAVE_GLO_PHY_SWH_L4_MY_014_007

Domain Resolution and grid Geographic coverage	The gridded product merges all available Significant Wave Height measurements over the global ocean. The maps are available daily at a 2-degree resolution for the VAVH_DAILY_MEAN field and 0.5-degree resolution for the VAVH_INST filed. The geographic coverage is constrained by the availability of wave measurements (see the geographic coverage for Level-3 SWH product above) and by the sea ice concentration (updated daily from CMEMS SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_001 product). The VAVH_INST fields utilise the wave measurements, accounting for their spatial and temporal proximity to provide wave information over each open ocean grid cell. The VAVH_DAILY fields are computed from the measurements available over one day (from 00UTC to 23:59UTC), filling only grid cells with measurements. See QUID for details.  Figure 2: *Example of gridded level-4 significant wave height maps (m) for the VAVH_INST field (top), estimated from all available missions Level 3 along-*

	track measurements, accounting for their spatial and temporal proximity and the VAVH_DAILY_MEAN field (bottom), computed from daily measurements.

II.3.2 WAVE-TAC SPC system

WAVE-TAC SPC SYSTEM	
Processed products	WAVE_GLO_PHY_SPC_L3_MY_014_006
Domain Resolution and grid Geographic coverage	The spectral measurements are delivered over the global area. They are based on L2 Sentinel-1A and Sentinel-1B wave measurements and available along each wave propagation path, from the generation storm to the land using a linear wave propagation model from observation time. The geographic coverage is constrained by the availability of wave measurements propagating away from land and islands and in the absence of ice (updated daily from CMEMS SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_001 product). Because of the absence of Sentinel-1 L2 wave acquisitions over the European coasts, this region is less well covered by this L3 products. The measurements are provided along the swell path over an irregular spatial grid with a regular time step (3h).  *Figure 3: Example of wave measurements from Level-3 Sentinel-1A and Sentinel-1B SARs given on February 5th 2018.*

[2]**See the QUID report for complete information about the processing**.

II.4 Processing information

II.4.1 Update Time

II.4.1.1 WAVE-TAC SWH system

Multi-year SWH products and their update frequency depend on reprocessings performed by external projects, such as the ESA CCI Sea State project.

II.4.1.2 WAVE-TAC SPC system

Multi-year SPC products are updated every 6 months.

The production chain runs over the last 7 months to produce level-3 SPC products from all latest available L2 measurements.

II.4.2 Temporal extent of data stored on delivery mechanism

II.4.2.1 WAVE-TAC SWH system

The temporal extent of the reprocessing covers the period 2002-2020.

II.4.2.2 WAVE-TAC SPC system

Each week, the swell field analysis is performed over the last 4 weeks. The older swell fields are unchanged from one week to the next as no recent observation can belong to the oldest swell fields. The last analyzed week, one month old, is thus ready to be incorporated in the multi-year product.

For each observation, the wave integral parameters are propagated from the storm source up to 10 days or before if reaching and dissipating at the coast.

II.4.3 Time averaging

II.4.3.1 WAVE-TAC SWH system

Instantaneous values are provided in L3 files (time is specified in the netCDF files). There is no temporal averaging nor interpolation performed, a low pass filter is applied to remove high frequency noise (see QUID for details).

In L4 files, an instantaneous value VAVH_INST is estimated from L3 measurements available within a time window around the date of the map (12 UTC). Daily statistics are also provided in the VAVH_DAILY fields (mean, standard deviation, maximum and number of measurements) and are computed using level 3 along-track measurements of the day from 00 UTC until 23:59 UTC.

II.4.3.2 WAVE-TAC SPC system

Instantaneous values are provided. There is no temporal averaging or interpolation performed in the L3 files.

III FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

IV.1 Nomenclature of files when downloaded through the CMEMS Web Portal Subset Service

Files nomenclature when downloaded through the CMEMS Web Portal Subset is based on product dataset name and a numerical reference related to the request date on the MIS.

The scheme is: **datasetname_nnnnnnnnnnnnn.nc**

where :

. datasetname is a character string within one of the following :

- cmems_obs-wave_glo_phy-swh_my_multi-l4-2deg_P1D_ (for SWH L4 products)

. nnnnnnnnnnnnn : 13 digit integer corresponding to the current time (download time) in milliseconds since January 1, 1970 midnight UTC.

.nc: standard NetCDF filename extension.

Example:

cmems_obs-wave_glo_phy-swh_my_multi-l4-2deg_P1D_<nnnnnnnnnnnnn>.nc

IV.2 Nomenclature of original files

All WAVE-TAC products (along-track level-3 and gridded level-4) are available through the CMEMS FTP Service and through the CMEMS Web Portal DGF Service.

The nomenclature of files downloaded is described below for the different wave products.

- WAVE_GLO_PHY_SWH_L3_MY_014_005 filenames:

global_vavh_l3_rep_<mission>_<datestart>_< dateend>_<dateprod>.nc

With:

- <mission>=j1, j2, j3, en, al, c2 or cfo
- < datestart >= date of the start of the 3-hour period YYYYmmddTHHMMSS
- <dateend>= date of the end of the 3-hour period YYYYmmddTHHMMSS

- <dateprod>=date of production of the file YYYYmmddTHHMMSS

- WAVE_GLO_PHY_SPC_L3_MY_014_006 filenames:

Swell field format:

dataset-wav-sar-l3-spc-rep-global-s1a_20210430T1200Z_20210523T1200Z_P20210517T1038Z_01-MAY-2021-18H_lon_W067_lat_S55

dataset-wav-sar-l3-spc-rep-global-<mission>_<datemin>_<datemax>_P<dateprod>_<storm_date>_lon_<storm_lon>_lat_<storm_lat>.nc

With:

- <mission>= s1a or s1b
- <datemin>= date of first measurement in the file YYYYmmddTHHMMZ
- <datemax>= date of last measurement in the file YYYYmmddTHHMMZ
- <dateprod>=date of production of the file YYYYmmddTHHMMZ
- <storm_date>=date of the generation storm source, dd-mmThhmmZ
- <storm_lon>=longitude of the storm source: E/WXXX [in degrees]
- <storm_lat>= latitude of the storm source:N/SXX [in degrees]

3-hour format:

dataset-wav-sar-l3-spc-rep-global-<mission>_<datemin>_<datemax>_P<dateprod>-3H.nc

With:

- <mission>= s1a or s1b
- <datemin>= date of first measurement in the file YYYYmmddTHHMMZ
- <datemax>= date of last measurement in the file YYYYmmddTHHMMZ
- <dateprod>=date of production of the file YYYYmmddTHHMMZ

- WAVE_GLO_PHY_SWH_L4_MY_014_007 filenames:

cmems_obs-wave_glo_phy-swh_my_multi_l4_2deg_P1D-<subfield>_< date>_P<dateprod>.nc

With:

- <subfield>= i for instantaneous and m for daily mean
- < date>= date of the daily wave field YYYYmmddT120000Z

- <dateprod>=date of production of the file YYYYmmddTHHMMSSZ

IV.3 Other information: mean centre of Products, land mask value, missing value

Real_Value = (Display_Value X scale_factor) + add_offset

Missing value, valid_min and valid_max values are to be compared to the raw netcdf data, before applying the scale_factor and add_offset.

The missing value for the SWH products is: -32767s (-999 for the number of measurements).

The missing value for the SPC products is: -999.

For gridded level-4 products, land and sea-ice masks are equal to "_FillValue".

IV.4 File size

DATASET NAME	NAME OF FILE	DIMENSION
cci_obs-wave_glo_phy-swh_my_<mission>-l3_PT1S **or** **cmems_obs-wave_glo_phy-swh_my_<mission>-l3_PT1S**	global_vavh_l3_rep_<mission>_<datestart>_<dateend>_<dateprod>.nc	Up to 200 Ko
dataset-wav-sar-l3-spc-rep-global-<mission>	dataset-wav-sar-l3-spc-rep-global-<mission>_<datemin>_<datemax>_P<dateprod>_<storm_date>_lon_<storm_lon>_lat_<storm_lat>.nc Or dataset-wav-sar-l3-spc-rep-global-<mission>_<datemin>_<datemax>_P<dateprod>-3H.nc	Up to 150 Mo for both

cmems_obs-wave_glo_phy-swh_my_multi-l4-2deg_P1D	cmems_obs-wave_glo_phy-swh_my_multi_l4_2deg_P1D-m_<date>_P<dateprod>.nc Or cmems_obs-wave_glo_phy-swh_my_multi_l4_0.5deg_P1D-i_<date>_P<dateprod>.nc	~100 Ko for daily_mean and ~1 Mo for inst

IV.5 Structure of files

Examples of the header of output NetCDF files are inserted in annex, for each dataset.

V REFERENCES

See the Quality Information Document (QUID): CMEMS-WAV-QUID-014-007-V2.0

VI ANNEX

VI.1 WAVE_GLO_PHY_SWH_L3_MY_014_005 files

An example of NetCDF metadata from Level-3 SWH is given below.

```
netcdf global_vavh_l3_rep_j3_20160228T060000_20160228T090000_20210910T140906 {
dimensions:
    time = 6507 ;
variables:
    short VAVH_UNFILTERED(time) ;
        VAVH_UNFILTERED:_FillValue = -32767s ;
        VAVH_UNFILTERED:scale_factor = 0.001 ;
        VAVH_UNFILTERED:units = "m" ;
        VAVH_UNFILTERED:comment = "Bias corrected. Original variable name in ESA CCI dataset: swh_adjusted" ;
        VAVH_UNFILTERED:valid_min = 0s ;
        VAVH_UNFILTERED:valid_max = 32767s ;
        VAVH_UNFILTERED:coordinates = "longitude latitude" ;
        VAVH_UNFILTERED:long_name = "Significant Wave Height on main altimeter frequency band" ;
        VAVH_UNFILTERED:standard_name = "sea_surface_wave_significant_height" ;
    int longitude(time) ;
        longitude:scale_factor = 1.e-06 ;
        longitude:units = "degrees_east" ;
        longitude:valid_min = 0 ;
        longitude:valid_max = 360000000 ;
        longitude:comment = "East longitude relative to Greenwich meridian." ;
        longitude:long_name = "longitude" ;
        longitude:standard_name = "longitude" ;
    short VAVH(time) ;
        VAVH:_FillValue = -32767s ;
        VAVH:scale_factor = 0.001 ;
        VAVH:units = "m" ;
        VAVH:comment = "Bias corrected and noise filtered. Original variable name in ESA CCI dataset: swh_denoised" ;
        VAVH:valid_min = 0s ;
        VAVH:valid_max = 32767s ;
        VAVH:coordinates = "longitude latitude" ;
        VAVH:long_name = "Significant Wave Height on main altimeter frequency band" ;
        VAVH:standard_name = "sea_surface_wave_significant_height" ;
    int latitude(time) ;
        latitude:scale_factor = 1.e-06 ;
        latitude:units = "degrees_north" ;
        latitude:valid_min = -90000000 ;
        latitude:valid_max = 90000000 ;
        latitude:comment = "Positive latitude is North latitude, negative latitude is South latitude." ;
        latitude:long_name = "latitude" ;
        latitude:standard_name = "latitude" ;
    double time(time) ;
        time:units = "seconds since 1981-01-01 00:00:00.0" ;
        time:calendar = "gregorian" ;
        time:axis = "T" ;
        time:long_name = "time (sec. since 1981-01-01)" ;
        time:standard_name = "time" ;

// global attributes:
        :title = "Multi-Year Jason-3 Global Ocean Along track significant wave height L3 product" ;
```

:comment = "These data are based on the ESA CCI Sea State project dataset and are formatted to follow CMEMS Level-3 SWH standards." ;
:summary = "Multi-Year Level-3 significant wave height measured by Jason-3 over the global ocean. Measurements are edited, cross-calibrated with other altimetry missions and reference in situ measurements." ;
:processing_level = "L3" ;
:institution = "CLS" ;
:creator_name = "Cersat" ;
:creator_url = "http://cersat.ifremer.fr" ;
:creator_email = "cersat@ifremer.fr" ;
:creator_institution = "Ifremer / Cersat" ;
:publisher_name = "CMEMS - Wave Thematic Assembly Center" ;
:publisher_url = "http://marine.copernicus.eu" ;
:publisher_email = "servicedesk.cmems@mercator-ocean.eu" ;
:references = "CCI Sea State Product Specification Document (PSD), v1.1 ; http://marine.copernicus.eu" ;
:contact = "servicedesk.cmems@mercator-ocean.eu" ;
:license = "These data are available free of charge under the CMEMS data policy, refer to http://marine.copernicus.eu/web/27-service-commitments-and-licence.php" ;
:project = "COPERNICUS MARINE ENVIRONMENT MONITORING SERVICE (CMEMS)" ;
:acknowledgement = "Please acknowledge the use of these data with the following statement: These data were provided by the ESA CCI Sea State project through the Copernicus Marine Service" ;
:Conventions = "CF-1.6" ;
:keywords = "Oceans > Ocean Topography > Significant Wave Height" ;
:keywords_vocabulary = "NetCDF COARDS Climate and Forecast Standard Names" ;
:Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
:standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention Standard Name Table v37" ;
:history = "2021-09-10T14:09:06: Formatting to CMEMS L3 standards\n2021-07-21T07:29:44: Creation by ESA CCI Sea State" ;
:date_created = "2021-07-21T07:29:44" ;
:date_modified = "2021-09-10T14:09:06" ;
:based_on = "ESA CCI Sea State L2P Jason-3 v2.0.6 dataset" ;
:cmems_product_id = "WAVE_GLO_PHY_SWH_L3_MY_014_005" ;
:product_version = "CMEMS_vDec2021" ;
:source = "CCI Sea State Jason-3 GDR to L2P Processor" ;
:source_version = "2.0.6" ;
:time_coverage_start = "2016-02-28T06:12:47.282980864" ;
:time_coverage_end = "2016-02-28T08:59:59.258629120" ;
:geospatial_lat_min = -80. ;
:geospatial_lat_max = 80. ;
:geospatial_lat_units = "degrees_north" ;
:geospatial_lon_min = 0. ;
:geospatial_lon_max = 360. ;
:geospatial_lon_units = "degrees_east" ;
:geospatial_vertical_units = "meters above mean sea level" ;
:geospatial_vertical_positive = "up" ;
:geospatial_vertical_min = 0. ;
:geospatial_vertical_max = 0. ;
:cdm_data_type = "swath" ;
:platform = "Jason-3" ;
}

VI.1.1 WAVE_GLO_PHY_SPC_L3_MY_014_006 files

The variables described in the netcdf files are organized in different ways depending on the level of information:

- The variables related to the propagated observations, these are given at the root of the netcdf file, as regular variables.

- One group for the observed variables, called "obs_params", in which are gathered variables related to the L2 observations (such as the spectral information).
- One group of variables dedicated to more advanced users, called "exp_params", in which are gathered variables related to the storm event at the origin of the L2 observations (such as the storm location, times, dimensions...).
- The gobal variables, in which are gathered more general parameters such as the algorithm version, the data producer, the first and last time of the observations and propagated observations.

Two examples of NetCDF metadata from these files are given below: one for the swell field format and one for the 3H format

VI.1.1.1 Swell field format

```
netcdf dataset-wav-sar-l3-spc-rep-global-s1a_20210427T0600Z_20210521T0600Z_P20210507T1255Z_27-APR-2021-06H_lon_W158_lat_N41.nc

{
dimensions:
obs = UNLIMITED ; // (1478 currently)
spectrum_wavenumber = 60 ;
spectrum_direction = 72 ;
propag_time = 193 ;
name_size = 67 ;
variables:
double time(propag_time) ;
time:units = "days since 1950-01-01T00:00:00Z" ;
time:long_name = "Date of propagation in CNES convention" ;
time:standard_name = "time" ;
float longitude(propag_time, obs) ;
longitude:_FillValue = -999.f ;
longitude:units = "degrees_east" ;
longitude:long_name = "Longitude of propagated observation" ;
longitude:standard_name = "longitude" ;
longitude:description = "East longitude relative to Greenwich meridian" ;
float latitude(propag_time, obs) ;
latitude:_FillValue = -999.f ;
latitude:units = "degrees_north" ;
latitude:long_name = "Latitude of propagated observation" ;
latitude:standard_name = "latitude" ;
latitude:description = "Positive latitude is North latitude, negative latitude is South latitude." ;
int VAVH(propag_time, obs) ;
VAVH:_FillValue = -999 ;
VAVH:units = "m" ;
VAVH:long_name = "Partition significant wave height along propagation" ;
VAVH:standard_name = "sea_surface_wave_significant_height" ;
VAVH:description = "The partition SWH is estimated along propagation before and after observations time, 3000 km from the storm source until it reaches land or propagates for too long. It uses decay and dissipation laws described in Ardhuin et al. 2009 and Collard et al. 2009" ;
VAVH:scale_factor = 0.001 ;
```

```
int VTPK(obs) ;
VTPK:units = "s" ;
VTPK:long_name = "Partition peak period" ;
VTPK:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum" ;
VTPK:description = "The partition peak period is assumed constant along the propagation (deep water). It is equals to the
partition peak period at observation time." ;
VTPK:scale_factor = 0.01 ;
int VPED(propag_time, obs) ;
VPED:_FillValue = -99999 ;
VPED:units = "degree" ;
VPED:long_name = "Partition peak direction along propagation (oceano. convention)" ;
VPED:standard_name = "sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
VPED:description = "The partition peak direction is estimated along propagation before and after observations time, from the
storm source until it reaches land or propagates for too long. It assumes great circle propagation. It corresponds to the
direction from which the waves arrive, in degrees from North, clockwise." ;
VPED:scale_factor = 0.1 ;

// global attributes:
:Processor_version = "1.5.10" ;
:Comment = "Spectral wave parameters along propagation and at observation time (measured by SAR) by swell field for
satellite mission Sentinel-1A" ;
:Conventions = "CF-1.6" ;
:creator_email = "servicedesk.cmems@mercator-ocean.eu" ;
:references = "http://marine.copernicus.eu" ;
:creator_url = "http://marine.copernicus.eu" ;
:project = "COPERNICUS MARINE ENVIRONMENT MONITORING SERVICE (CMEMS)" ;
:contact = "servicedesk.cmems@mercator-ocean.eu" ;
:processing_level = "L3" ;
:creation_date = "20210507T1255Z" ;
:keywords_vocabulary = "NetCDF COARDS Climate and Forecast Standard Names" ;
:institution = "CLS" ;
:creator_name = "CMEMS - Wave Thematic Assembly Center" ;
:license = "http://marine.copernicus.eu/web/27-service-commitments-and-licence.php" ;
:standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention Standard Name Table v37" ;
:title = "NRT Sentinel-1A Global Ocean Spectral Wave L3 product" ;
:source_platform = "Sentinel-1A measurements" ;
:based_on = "Sentinel-1A WV_L2_OCN" ;
:first_obs_time = "20210426T062825Z" ;
:last_obs_time = "20210506T063317Z" ;
:first_propag_time = "20210427T0600Z" ;
:last_propag_time = "20210521T0600Z" ;
:Rough_storm_source_point_longitude = " 161.161 deg" ;
:Rough_storm_source_point_latitude = "  38.739 deg" ;
:Rough_storm_source_size = "1259.908 km" ;
:Swell_generation_duration = "1.75 days" ;

group: obs_params {
  variables:
        double VAVH(obs) ;
                VAVH:units = "m" ;
                VAVH:long_name = "Partition significant wave height at observation time" ;
                VAVH:standard_name = "sea_surface_wave_significant_height" ;
                VAVH:description = "The partition SWH is retrieved from the L2 OCN WV products." ;
        float VPED(obs) ;
                VPED:units = "degree" ;
                VPED:long_name = "Partition peak direction at observation time (oceano. convention)" ;
                VPED:standard_name = "sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
```

```
                VPED:description = "The partition peak direction is retrieved from the L2 OCN WV products. It
corresponds to the direction from which the waves arrive, in degrees from North, clockwise." ;
        float VTPK(obs) ;
                VTPK:units = "s" ;
                VTPK:long_name = "Partition peak period" ;
                VTPK:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum" ;
                VTPK:description = "The partition peak period is retrieved from the L2 OCN WV products." ;
        double time(obs) ;
                time:units = "days since 1950-01-01T00:00:00Z" ;
                time:long_name = "Observation time" ;
                time:standard_name = "obs_time" ;
        float longitude(obs) ;
                longitude:units = "degrees_east" ;
                longitude:long_name = "Longitude of observation" ;
                longitude:standard_name = "obs_longitude" ;
                longitude:description = "East longitude relative to Greenwich meridian" ;
        float latitude(obs) ;
                latitude:units = "degrees_north" ;
                latitude:long_name = "Latitude of observation" ;
                latitude:standard_name = "obs_latitude" ;
                latitude:description = "Positive latitude is North latitude, negative latitude is South latitude." ;
        float wave_spec(spectrum_wavenumber, spectrum_direction, obs) ;
                wave_spec:units = "m^4" ;
                wave_spec:long_name = "Estimated wave spectrum on log-polar grid" ;
                wave_spec:standard_name = "wave_spectrum" ;
        int partition_domain_spec(spectrum_wavenumber, spectrum_direction, obs) ;
                partition_domain_spec:units = "1" ;
                partition_domain_spec:long_name = "Estimated wave partition spectral domain on log-polar grid" ;
                partition_domain_spec:standard_name = "wave_partition" ;
        float wavenumber_spec(spectrum_wavenumber) ;
                wavenumber_spec:units = "rad/m" ;
                wavenumber_spec:long_name = "Array of the wavenumbers for polar spectra" ;
                wavenumber_spec:standard_name = "wavenumber_spec" ;
        float direction_spec(spectrum_direction) ;
                direction_spec:units = "degree" ;
                direction_spec:long_name = "Array of angular values for polar spectra (degrees from North, clockwise)" ;
                direction_spec:standard_name = "direction_spec" ;
        float nv(obs) ;
                nv:units = "1" ;
                nv:long_name = "Normalised variance estimated in the input L2 SLC image" ;
        float inten(obs) ;
                inten:units = "1" ;
                inten:long_name = "Input SLC image intensity estimated in the input L2 SLC image" ;
        float snr(obs) ;
                snr:units = "dB" ;
                snr:long_name = "Signal to noise ratio estimated in the input L2 cross-spectrum" ;
        float resolution_spec(spectrum_direction, obs) ;
                resolution_spec:units = "m" ;
                resolution_spec:long_name = "Azimuth cut-off wavelength as function of wave direction, for the input L2
spectrum" ;
                resolution_spec:standard_name = "resolution_spec" ;
        int VPED_flag(obs) ;
                VPED_flag:units = "1" ;
                VPED_flag:long_name = "Global quality flag for each L2 swell partition (partition peak direction coherency)
- Values from 0 to 5" ;
                VPED_flag:standard_name =
"sea_surface_wave_from_direction_at_variance_spectral_density_maximum_flag" ;
```

```
                VPED_flag:description = "The quality of each L2 swell partition peak direction is examined through 5
successive filters. Data quality goes from 0 (poor) to 5 (passed all the filters)." ;
        int VTPK_flag(obs) ;
                VTPK_flag:units = "1" ;
                VTPK_flag:long_name = "Global quality flag for each L2 swell partition (partition peak period coherency) -
Values from 0 to 5" ;
                VTPK_flag:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_flag" ;
                VTPK_flag:description = "The quality of each L2 swell partition peak period is examined through 5
successive filters. Data quality goes from 0 (poor) to 5 (passed all the filters)." ;
        int VAVH_flag(obs) ;
                VAVH_flag:units = "1" ;
                VAVH_flag:long_name = "Global Hs quality flag for each L2 swell partition (partition Hs coherency) -
Values from 0 to 1" ;
                VAVH_flag:standard_name = "sea_surface_wave_significant_height_flag" ;
                VAVH_flag:description = "The quality of each L2 swell partition Hs is examined through successive filters.
Data quality goes from 0 (poor) to 1 (passed all the filters)." ;
        int n_posneg(obs) ;
                n_posneg:_FillValue = -1 ;
                n_posneg:units = "1" ;
                n_posneg:long_name = "L2 ncdf posneg (only for SWIM source)" ;
                n_posneg:standard_name = "n_posneg" ;
        int n_box(obs) ;
                n_box:_FillValue = -1 ;
                n_box:units = "1" ;
                n_box:long_name = "L2 ncdf box (only for SWIM source)" ;
                n_box:standard_name = "n_posneg" ;
        char L2_source_product(obs, name_size) ;
                L2_source_product:long_name = "L2 ncdf source product filename" ;
                L2_source_product:standard_name = "L2_source_product" ;
        byte L2_partition_quality_flag(obs) ;
                L2_partition_quality_flag:_FillValue = -128b ;
                L2_partition_quality_flag:long_name = "Quality flag per partition for each L2 swell partition" ;
                L2_partition_quality_flag:valid_range = 0b, 4b ;
                L2_partition_quality_flag:flag_values = 0b, 1b, 2b, 3b, 4b ;
                L2_partition_quality_flag:flag_meanings = "very_good good medium low poor" ;
 } // group obs_params

group: exp_params {
 variables:
        float foc_longitude(obs) ;
                foc_longitude:units = "degrees_east" ;
                foc_longitude:long_name = "Focal longitude" ;
                foc_longitude:standard_name = "foc_longitude" ;
                foc_longitude:description = "The focal longitude corresponds to the center longitude of all the propagated
observations that are refocusing at the same time step. It can differ among the set of propagated observations." ;
        float foc_latitude(obs) ;
                foc_latitude:units = "degrees_north" ;
                foc_latitude:long_name = "Focal latitude" ;
                foc_latitude:standard_name = "foc_latitude" ;
                foc_latitude:description = "The focal latitude corresponds to the center latitude of all the propagated
observations that are refocusing at the same time step. It can differ among the set of propagated observations." ;
        float foc_size(obs) ;
                foc_size:units = "km" ;
                foc_size:long_name = "Focal size diameter" ;
                foc_size:standard_name = "foc_size" ;
                foc_size:description = "The focal size diameter corresponds to the diameter of the refocusing region
where all the propagated observations that are refocusing at the same time step have converged. It can differ among the set
of propagated observations." ;
```

```
        float foc_size_along(obs) ;
                foc_size_along:units = "km" ;
                foc_size_along:long_name = "Focal size diameter projected in main propagation direction" ;
                foc_size_along:standard_name = "foc_size_along" ;
                foc_size_along:description = "Idem but projected in main propagation direction. It can differ among the
set of propagated observations." ;
        float foc_size_across(obs) ;
                foc_size_across:units = "km" ;
                foc_size_across:long_name = "Focal size diameter projected in main transverse propagation direction" ;
                foc_size_across:standard_name = "foc_size_across" ;
                foc_size_across:description = "Idem but projected in main transverse direction. It can differ among the set
of propagated observations." ;
        double foc_time(obs) ;
                foc_time:units = "days since 1950-01-01T00:00:00Z" ;
                foc_time:long_name = "Refocusing time in CNES convention" ;
                foc_time:standard_name = "foc_time" ;
                foc_time:description = "The refocusing time corresponds to the time when the propagated observations
are converging. It can differ among the set of propagated observations." ;
  } // group exp_params
}
```

VI.1.1.2 3H format

netcdf dataset-wav-sar-l3-spc-nrt-global-s1a_20210526T150000Z_20210526T180000Z_P20210608T0410Z-3H-nrt

```
{
dimensions:
obs = 183 ;
spectrum_wavenumber = 60 ;
spectrum_direction = 72 ;
propag_time = 184 ;
name_size = 67 ;
storm_size = 32 ;
variables:
int VAVH(propag_time, obs) ;
VAVH:_FillValue = -999 ;
VAVH:units = "Meters" ;
VAVH:long_name = "Partition Significant Wave Height along propagation" ;
VAVH:description = "The partition SWH is estimated along propagation before and after observations time, 4000 km from the
storm source until it reaches land or propagates for too long. It uses decay and dissipation laws described in Ardhuin et al. 2009
and Collard et al. 2009" ;
VAVH:standard_name = "sea_surface_wave_significant_height" ;
VAVH:scale_factor = 0.001 ;
int VPED(propag_time, obs) ;
VPED:_FillValue = -99999 ;
VPED:units = "Degrees" ;
VPED:long_name = "Partition Peak Direction along propagation (met. convention)" ;
VPED:description = "The partition peak direction is estimated along propagation before and after observations time, from the
storm source until it reaches land or propagates for too long. It assumes great circle propagation. It corresponds to the direction
from which the waves arrive, in degrees from North, clockwise." ;
VPED:standard_name = "sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
VPED:scale_factor = 0.1 ;
float latitude(propag_time, obs) ;
latitude:_FillValue = -999.f ;
latitude:units = "degrees_north" ;
latitude:long_name = "latitude of propagated observation" ;
latitude:description = "Positive latitude is North latitude, negative latitude is South latitude." ;
latitude:standard_name = "latitude" ;
```

```
int VTPK(obs) ;
VTPK:units = "Seconds" ;
VTPK:long_name = "Partition Peak Period" ;
VTPK:description = "The partition peak period is assumed constant along the propagation (deep water). It is equals to the partition peak period at observation time." ;
VTPK:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum" ;
VTPK:scale_factor = 0.01 ;
float longitude(propag_time, obs) ;
longitude:_FillValue = -999.f ;
longitude:units = "degrees_east" ;
longitude:long_name = "longitude of propagated observation" ;
longitude:description = "East longitude relative to Greenwich meridian" ;
longitude:standard_name = "longitude" ;
double time(propag_time) ;
time:units = "days since 1950-01-01T00:00:00Z" ;
time:long_name = "Date of propagation in CNES convention" ;

// global attributes:
:Processor_version = "1.3.8" ;
:source_platform = "Sentinel-1A measurements" ;
:last_obs_time = "20210526T163200Z" ;
:license = "http://marine.copernicus.eu/web/27-service-commitments-and-licence.php" ;
:creator_name = "CMEMS - Wave Thematic Assembly Center" ;
:title = "NRT Sentinel-1A Global Ocean Spectral Wave L3 product" ;
:creator_email = "servicedesk.cmems@mercator-ocean.eu" ;
:standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention Standard Name Table v37" ;
:last_propag_time = "20210606T030000Z" ;
:Conventions = "CF-1.6" ;
:project = "COPERNICUS MARINE ENVIRONMENT MONITORING SERVICE (CMEMS)" ;
:keywords_vocabulary = "NetCDF COARDS Climate and Forecast Standard Names" ;
:contact = "servicedesk.cmems@mercator-ocean.eu" ;
:references = "http://marine.copernicus.eu" ;
:first_obs_time = "20210526T150719Z" ;
:based_on = "Sentinel-1A WV_L2_OCN" ;
:creator_url = "http://marine.copernicus.eu" ;
:first_propag_time = "20210514T060000Z" ;
:institution = "CLS" ;
:processing_level = "L3" ;
:Comment = "Spectral wave parameters along propagation and at observation time (measured by SAR) for satellite mission s1a" ;

group: obs_params {
  variables:
        float VPED(obs) ;
                VPED:units = "Degrees" ;
                VPED:long_name = "Partition Peak Direction at observation time (met. convention)" ;
                VPED:description = "The partition peak direction is retrieved from the L2 OCN WV products. It corresponds to the direction from which the waves arrive, in degrees from North, clockwise." ;
                VPED:standard_name = "sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
        float snr(obs) ;
                snr:units = "dB" ;
                snr:long_name = "Signal to noise ratio estimated in the input L2 cross-spectrum" ;
        float VTPK(obs) ;
                VTPK:units = "Seconds" ;
                VTPK:long_name = "Partition Peak Period" ;
                VTPK:description = "The partition peak period is retrieved from the L2 OCN WV products." ;
                VTPK:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum" ;
        int partition_domain_spec(spectrum_wavenumber, spectrum_direction, obs) ;
```

```
                partition_domain_spec:units = "1" ;
                partition_domain_spec:long_name = "Estimated wave partition spectral domain on log-polar grid" ;
                partition_domain_spec:standard_name = "wave_partition" ;
        int VPED_flag(obs) ;
                VPED_flag:units = "1" ;
                VPED_flag:long_name = "Global quality flag for each L2 swell partition (partition peak direction coherency) - Values from 0 to 5" ;
                VPED_flag:description = "The quality of each L2 swell partition peak direction is examined through 5 successive filters. Data quality goes from 0 (poor) to 5 (passed all the filters)." ;
                VPED_flag:standard_name = "sea_surface_wave_from_direction_at_variance_spectral_density_maximum_flag" ;
        byte L2_partition_quality_flag(obs) ;
                L2_partition_quality_flag:_FillValue = -128b ;
                L2_partition_quality_flag:long_name = "Quality flag per partition for each L2 swell partition" ;
                L2_partition_quality_flag:valid_range = 0b, 4b ;
                L2_partition_quality_flag:flag_values = 0b, 1b, 2b, 3b, 4b ;
                L2_partition_quality_flag:flag_meanings = "very_good good medium low poor" ;
        char L2_source_product(obs, name_size) ;
                L2_source_product:long_name = "L2 ncdf source product filename" ;
                L2_source_product:standard_name = "L2_source_product" ;
        float direction_spec(spectrum_direction) ;
                direction_spec:units = "degrees from North, clockwise " ;
                direction_spec:long_name = "Array of angular values for polar spectra" ;
                direction_spec:standard_name = "direction_spec" ;
        float longitude(obs) ;
                longitude:units = "degrees_east" ;
                longitude:long_name = "longitude of observation" ;
                longitude:description = "East longitude relative to Greenwich meridian" ;
                longitude:standard_name = "obs_longitude" ;
        float inten(obs) ;
                inten:units = "1" ;
                inten:long_name = "Input SLC image intensity estimated in the input L2 SLC image" ;
        int VAVH_flag(obs) ;
                VAVH_flag:units = "1" ;
                VAVH_flag:long_name = "Global Hs quality flag for each L2 swell partition (partition Hs coherency) - Values from 0 to 1" ;
                VAVH_flag:description = "The quality of each L2 swell partition Hs is examined through successive filters. Data quality goes from 0 (poor) to 1 (passed all the filters)." ;
                VAVH_flag:standard_name = "sea_surface_wave_significant_height_flag" ;
        float wavenumber_spec(spectrum_wavenumber) ;
                wavenumber_spec:units = "rad/m" ;
                wavenumber_spec:long_name = "Array of the wavenumbers for polar spectra" ;
                wavenumber_spec:standard_name = "wavenumber_spec" ;
        char storm_id(obs, storm_size) ;
                storm_id:long_name = "storm identification name" ;
                storm_id:description = "The storm id that can be use to find the associated file dataset for a particular storm." ;
                storm_id:standard_name = "storm_id" ;
        float latitude(obs) ;
                latitude:units = "degrees_north" ;
                latitude:long_name = "latitude of observation" ;
                latitude:description = "Positive latitude is North latitude, negative latitude is South latitude." ;
                latitude:standard_name = "obs_latitude" ;
        float nv(obs) ;
                nv:units = "1" ;
                nv:long_name = "Normalised variance estimated in the input L2 SLC image" ;
        int VTPK_flag(obs) ;
                VTPK_flag:units = "1" ;
```

```
                VTPK_flag:long_name = "Global quality flag for each L2 swell partition (partition peak period coherency) - Values from 0 to 5" ;
                VTPK_flag:description = "The quality of each L2 swell partition peak period is examined through 5 successive filters. Data quality goes from 0 (poor) to 5 (passed all the filters)." ;
                VTPK_flag:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_flag" ;
        double VAVH(obs) ;
                VAVH:units = "Meters" ;
                VAVH:long_name = "Partition Significant Wave Height at observation time" ;
                VAVH:description = "The partition SWH is retrieved from the L2 OCN WV products." ;
                VAVH:standard_name = "sea_surface_wave_significant_height" ;
        double time(obs) ;
                time:units = "days since 1950-01-01T00:00:00Z" ;
                time:long_name = "Observation Time in CNES convention" ;
                time:standard_name = "obs_time" ;
        float resolution_spec(spectrum_direction, obs) ;
                resolution_spec:units = "m" ;
                resolution_spec:long_name = "Azimuth cut-off wavelength as function of wave direction, for the input L2 spectrum" ;
                resolution_spec:standard_name = "resolution_spec" ;
        float wave_spec(spectrum_wavenumber, spectrum_direction, obs) ;
                wave_spec:units = "m^4" ;
                wave_spec:long_name = "Estimated wave spectrum on log-polar grid" ;
                wave_spec:standard_name = "wave_spectrum" ;
 } // group obs_params

group: exp_params {
 variables:
        float foc_size(obs) ;
                foc_size:units = "Km" ;
                foc_size:long_name = "Focal size diameter" ;
                foc_size:description = "The focal size diameter corresponds to the diameter of the refocusing region where all the propagated observations that are refocusing at the same time step have converged. It can differ among the set of propagated observations." ;
                foc_size:standard_name = "foc_size" ;
        float foc_latitude(obs) ;
                foc_latitude:units = "degrees_north" ;
                foc_latitude:long_name = "Focal latitude" ;
                foc_latitude:description = "The focal latitude corresponds to the center latitude of all the propagated observations that are refocusing at the same time step. It can differ among the set of propagated observations." ;
                foc_latitude:standard_name = "foc_latitude" ;
        double foc_time(obs) ;
                foc_time:units = "days since 1950-01-01T00:00:00Z" ;
                foc_time:long_name = "Refocusing time in CNES convention" ;
                foc_time:description = "The refocusing time corresponds to the time when the propagated observations are converging. It can differ among the set of propagated observations." ;
                foc_time:standard_name = "foc_time" ;
        float foc_size_along(obs) ;
                foc_size_along:units = "Km" ;
                foc_size_along:long_name = "Focal size diameter projected in main propagation direction" ;
                foc_size_along:description = "Idem but projected in main propagation direction. It can differ among the set of propagated observations." ;
                foc_size_along:standard_name = "foc_size_along" ;
        float foc_longitude(obs) ;
                foc_longitude:units = "degrees_east" ;
                foc_longitude:long_name = "Focal longitude" ;
                foc_longitude:description = "The focal longitude corresponds to the center longitude of all the propagated observations that are refocusing at the same time step. It can differ among the set of propagated observations." ;
                foc_longitude:standard_name = "foc_longitude" ;
```

```
            float foc_size_across(obs) ;
                        foc_size_across:units = "Km" ;
                        foc_size_across:long_name = "Focal size diameter projected in main transverse propagation direction" ;
                        foc_size_across:description = "Idem but projected in main transverse direction. It can differ among the set
of propagated observations." ;
                        foc_size_across:standard_name = "foc_size_across" ;
 } // group exp_params
}
```

VI.1.2 WAVE_GLO_PHY_SWH_L4_MY_014_007 files

Examples of the two NetCDF files from Level-4 SWH are given below.

```
cmems_obs-wave_glo_phy-swh_my_multi_l4_2deg_P1D-m_20200818T120000Z_P20210901T070007Z.nc
netcdf cmems_obs-wave_glo_phy-swh_my_multi_l4_2deg_P1D-m_20200818T120000Z_P20210901T070007Z {
dimensions:
    time = 1 ;
    latitude = 90 ;
    longitude = 180 ;
    nv = 2 ;
variables:
    float time(time) ;
        time:axis = "T" ;
        time:long_name = "time" ;
        time:standard_name = "time" ;
        time:units = "days since 1950-01-01" ;
        time:calendar = "gregorian" ;
    float latitude(latitude) ;
        latitude:axis = "Y" ;
        latitude:bounds = "lat_bnds" ;
        latitude:long_name = "latitude" ;
        latitude:standard_name = "latitude" ;
        latitude:units = "degrees_north" ;
        latitude:valid_max = 89. ;
        latitude:valid_min = -89. ;
    float longitude(longitude) ;
        longitude:axis = "X" ;
        longitude:bounds = "lon_bnds" ;
        longitude:long_name = "longitude" ;
        longitude:standard_name = "longitude" ;
        longitude:units = "degrees_east" ;
        longitude:valid_max = 179. ;
        longitude:valid_min = -179. ;
    int nv(nv) ;
        nv:comment = "Vertex" ;
        nv:units = "1" ;
    float lat_bnds(latitude, nv) ;
        lat_bnds:comment = "latitude values at the north and south bounds of each pixel." ;
        lat_bnds:units = "degrees_north" ;
    float lon_bnds(longitude, nv) ;
        lon_bnds:comment = "longitude values at the west and east bounds of each pixel." ;
        lon_bnds:units = "degrees_east" ;

    short VAVH_DAILY_MEAN(time, latitude, longitude) ;
        VAVH_DAILY_MEAN:_FillValue = -32767s ;
        VAVH_DAILY_MEAN:ancillary_variables = "VAVH_DAILY_NBR" ;
```

```
        VAVH_DAILY_MEAN:cell_methods = "area: mean" ;
        VAVH_DAILY_MEAN:comment = "Daily statistics are computed from all available altimetry missions Level 3 along-
track measurements from 00 UTC until 23:59 UTC." ;
        VAVH_DAILY_MEAN:coordinates = "longitude latitude" ;
        VAVH_DAILY_MEAN:long_name = "Daily mean significant wave height" ;
        VAVH_DAILY_MEAN:standard_name = "sea_surface_wave_significant_height" ;
        VAVH_DAILY_MEAN:units = "m" ;
        VAVH_DAILY_MEAN:valid_max = 32767 ;
        VAVH_DAILY_MEAN:valid_min = 0 ;
        VAVH_DAILY_MEAN:scale_factor = 0.001 ;
    short VAVH_DAILY_STD(time, latitude, longitude) ;
        VAVH_DAILY_STD:_FillValue = -32767s ;
        VAVH_DAILY_STD:ancillary_variables = "VAVH_DAILY_NBR" ;
        VAVH_DAILY_STD:cell_methods = "area: standard_deviation" ;
        VAVH_DAILY_STD:comment = "Daily statistics are computed from all available altimetry missions Level 3 along-track
measurements from 00 UTC until 23:59 UTC." ;
        VAVH_DAILY_STD:coordinates = "longitude latitude" ;
        VAVH_DAILY_STD:long_name = "Daily standard deviation of significant wave height" ;
        VAVH_DAILY_STD:standard_name = "sea_surface_wave_significant_height" ;
        VAVH_DAILY_STD:units = "m" ;
        VAVH_DAILY_STD:valid_max = 32767 ;
        VAVH_DAILY_STD:valid_min = 0 ;
        VAVH_DAILY_STD:scale_factor = 0.001 ;
    short VAVH_DAILY_MAX(time, latitude, longitude) ;
        VAVH_DAILY_MAX:_FillValue = -32767s ;
        VAVH_DAILY_MAX:ancillary_variables = "VAVH_DAILY_NBR" ;
        VAVH_DAILY_MAX:cell_methods = "area: maximum" ;
        VAVH_DAILY_MAX:comment = "Daily statistics are computed from all available altimetry missions Level 3 along-
track measurements from 00 UTC until 23:59 UTC." ;
        VAVH_DAILY_MAX:coordinates = "longitude latitude" ;
        VAVH_DAILY_MAX:long_name = "Daily maximum significant wave height" ;
        VAVH_DAILY_MAX:standard_name = "sea_surface_wave_significant_height" ;
        VAVH_DAILY_MAX:units = "m" ;
        VAVH_DAILY_MAX:valid_max = 32767 ;
        VAVH_DAILY_MAX:valid_min = 0 ;
        VAVH_DAILY_MAX:scale_factor = 0.001 ;
    int VAVH_DAILY_NBR(time, latitude, longitude) ;
        VAVH_DAILY_NBR:_FillValue = -999 ;
        VAVH_DAILY_NBR:comment = "Daily statistics are computed from all available altimetry missions Level 3 along-track
measurements from 00 UTC until 23:59 UTC." ;
        VAVH_DAILY_NBR:coordinates = "longitude latitude" ;
        VAVH_DAILY_NBR:long_name = "Daily number of significant wave height measurements" ;
        VAVH_DAILY_NBR:standard_name = "sea_surface_wave_significant_height number_of_observations" ;
        VAVH_DAILY_NBR:units = "1" ;

// global attributes:
        :Conventions = "CF-1.6" ;
        :Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
        :cdm_data_type = "Grid" ;
        :comment = "Significant Wave Height measured by altimetry" ;
        :contact = "servicedesk.cmems@mercator-ocean.eu" ;
        :creator_email = "servicedesk.cmems@mercator-ocean.eu" ;
        :creator_name = "CMEMS - Wave Thematic Assembly Center" ;
        :creator_url = "http://marine.copernicus.eu" ;
        :date_created = "2021-09-01T07:00:07Z" ;
        :geospatial_lat_max = 89. ;
        :geospatial_lat_min = -89. ;
        :geospatial_lat_resolution = 2 ;
```

```
        :geospatial_lat_units = "degrees_north" ;
        :geospatial_lon_max = 179. ;
        :geospatial_lon_min = -179. ;
        :geospatial_lon_resolution = 2 ;
        :geospatial_lon_units = "degrees_east" ;
        :history = "2021-09-01T07:00:07Z: Creation" ;
        :institution = "CLS" ;
        :keywords = "Oceans > Ocean Topography > significant wave height" ;
        :keywords_vocabulary = "NetCDF COARDS Climate and Forecast Standard Names" ;
        :license = "http://marine.copernicus.eu/web/27-service-commitments-and-licence.php" ;
        :platform = "CFOSAT, Jason-3, SARAL-Altika" ;
        :processing_level = "L4" ;
        :product_version = "CMEMS vDec2021" ;
        :cmems_product_id = "WAVE_GLO_PHY_SWH_L4_MY_014_007" ;
        :project = "COPERNICUS MARINE ENVIRONMENT MONITORING SERVICE (CMEMS)" ;
        :references = "http://marine.copernicus.eu" ;
        :software_version = "L4RT_SWH: 1.2.0" ;
        :source = "Altimetry measurements" ;
        :standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention Standard Name Table v37" ;
        :summary = "Multi-Year Level-4 Significant Wave Height and derived variables measured by multi-satellite altimetry
observations over Global Ocean." ;
        :time_coverage_duration = "P1D" ;
        :time_coverage_end = "2020-08-18T12:00:00Z" ;
        :time_coverage_resolution = "P1D" ;
        :time_coverage_start = "2020-08-18T12:00:00Z" ;
        :title = "Multi-Year merged all satellites Global Ocean Gridded Significant Wave Height L4 product and mean
variables" ;
}

cmems_obs-wave_glo_phy-swh_my_multi_l4_0.5deg_P1D-i_20071222T120000Z_P20240528T000000Z
netcdf  cmems_obs-wave_glo_phy-swh_my_multi_l4_0.5deg_P1D-i _20071222T120000Z_P20240528T000000Z {
dimensions:
    longitude = 720 ;
    latitude = 361 ;
    time = 1 ;
    nv = 2 ;
variables:
    double longitude(longitude) ;
        longitude:_FillValue = NaN ;
        longitude:bounds = "longitude_bounds" ;
        longitude:long_name = "longitude" ;
        longitude:axis = "X" ;
        longitude:standard_name = "longitude" ;
        longitude:units = "degrees_east" ;
        longitude:valid_max = 179.5 ;
        longitude:valid_min = -180. ;
    double latitude(latitude) ;
        latitude:long_name = "latitude" ;
        latitude:valid_max = 90. ;
        latitude:_FillValue = NaN ;
        latitude:axis = "Y" ;
        latitude:standard_name = "latitude" ;
        latitude:bounds = "latitude_bounds" ;
        latitude:units = "degrees_north" ;
        latitude:valid_min = -90. ;
    double time(time) ;
        time:long_name = "time" ;
        time:calendar = "gregorian" ;
```

```
        time:_FillValue = NaN ;
        time:axis = "T" ;
        time:standard_name = "time" ;
        time:units = "days since 1950-01-01 00:00:00" ;
    double lat_bnds(latitude, nv) ;
        lat_bnds:_FillValue = NaN ;
        lat_bnds:comment = "latitude values at the west and east bounds of each pixel." ;
        lat_bnds:units = "degrees_north" ;
    double lon_bnds(longitude, nv) ;
        lon_bnds:_FillValue = NaN ;
        lon_bnds:comment = "longitude values at the west and east bounds of each pixel." ;
        lon_bnds:units = "degrees_east" ;
    double VAVH_INST(time, latitude, longitude) ;
        VAVH_INST:long_name = "Instantaneous significant wave height" ;
        VAVH_INST:valid_max = 32767LL ;
        VAVH_INST:_FillValue = NaN ;
        VAVH_INST:standard_name = "sea_surface_wave_significant_height" ;
        VAVH_INST:comment = "The gridded instantaneous significant wave height is estimated from all available altimetry
missions Level 3 along-track measurements, accounting for their spatial and temporal proximity." ;
        VAVH_INST:units = "m" ;
        VAVH_INST:valid_min = 0LL ;
    int64 VAVH_INST_FLAG(time, latitude, longitude) ;
        VAVH_INST_FLAG:comment = "Flag associated with a post-processing modification of the data (threshold set to 0
m). Where the value was invalid, the flag is set to 1, otherwise to 0." ;
        VAVH_INST_FLAG:long_name = "Flag associated with the modification of VAVH_INST variable." ;
        VAVH_INST_FLAG:standard_name = "sea_surface_wave_significant_height_flag" ;
        VAVH_INST_FLAG:units = "-" ;
        VAVH_INST_FLAG:valid_range = 0LL, 1LL ;

// global attributes:
        :Conventions = "CF-1.6" ;
        :Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
        :cdm_data_type = "Grid" ;
        :cmems_product_id = "WAVE_GLO_PHY_SWH_L4_MY_014_007" ;
        :comment = "Significant Wave Height measured by altimetry" ;
        :contact = "servicedesk.cmems@mercator-ocean.eu" ;
        :creator_name = "CMEMS - Wave Thematic Assembly Center" ;
        :creator_url = "http://marine.copernicus.eu" ;
        :date_created = "2024-05-28T00:00:00Z" ;
        :geospatial_lat_max = 90. ;
        :geospatial_lat_min = -90. ;
        :geospatial_lat_resolution = 0.5 ;
        :geospatial_lat_units = "degrees_north" ;
        :geospatial_lon_max = 179.5 ;
        :geospatial_lon_min = -180. ;
        :geospatial_lon_resolution = 0.5 ;
        :geospatial_lon_units = "degrees_east" ;
        :history = "2024-05-28T00:00:00Z: Creation" ;
        :institution = "CLS" ;
        :keywords = "Oceans > Ocean Topography > significant wave height" ;
        :keywords_vocabulary = "NetCDF COARDS Climate and Forecast Standard Names" ;
        :license = "http://marine.copernicus.eu/web/27-service-commitments-and-licence.php" ;
        :platform = "Jason-1, Envisat" ;
        :processing_level = "L4" ;
        :product_version = "202112" ;
        :project = "COPERNICUS MARINE ENVIRONMENT MONITORING SERVICE (CMEMS)" ;
        :references = "http://marine.copernicus.eu" ;
        :software_version = "L4RT_SWH: DT_1.2.1" ;
```

:source = "Altimetry measurements" ;
:standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention Standard Name Table v37" ;
:summary = "Multi-Year Level-4 Significant Wave Height and derived variables measured by multi-satellite altimetry observations over Global Ocean." ;
:time_coverage_duration = "P1D" ;
:time_coverage_end = "2007-12-22T12:00:00Z" ;
:time_coverage_resolution = "P1D" ;
:time_coverage_start = "2007-12-22T12:00:00Z" ;
:title = "Multi-Year merged all satellites Global Ocean Gridded Significant Wave Height L4 product and instantaneous variables" ;
:coordinates = "lat_bnds lon_bnds" ;
}